Ad hoc semantic reconciliation by cognitive agents: a four-setting synthesis

Programme abstract — Two software systems that must exchange information rarely share a model of the world. The classical answer is to agree a common standard in advance; this programme asks whether two systems that can reason can instead reconcile their divergent models for the occasion — ad hoc, machine-to-machine, with no standard settled beforehand. Across four operational settings in network and service management — configuration, intent, cross-domain provisioning, and observability — it builds a measured harness, scores every reconciliation against a validated gold standard, and varies one master control: where the cognition sits, from two live reasoning agents through one inert to both inert. The through-line the four settings establish is a single thesis. It is cognition that completes a reconciliation. Descriptor methods — matching names, then names plus a gloss — carry it to a ceiling and stop; where both systems can reason, the remainder is closed autonomously, with no agreed model and no human in the loop. A thin, published reference can substitute for cognition or supply an inert side the information it lacks, but it reaches information and stops at authority. And the pragmatic layer — what a reconciled thing is for, whether it matters, who decides — is the frontier the descriptor methods never reach, decisive for meaning and itself bounded by the agent's capability. This document is the synthesis that sits atop the four setting reports: it introduces the concepts once, distils each setting to its essentials, and gathers the findings so they can be read as one result. The full evidence for any setting lives in its own report (1/4–4/4), to which this one points throughout.

Part I — The idea

1. The problem: two models, and no time to standardise

When two software systems must work together but hold different models — of the same domain, or of adjacent domains that must connect — someone has to reconcile those models. The scope a model must cover is not fixed, the useful level of abstraction depends on the task, and capable engineers, working independently, produce different but defensible models of the same network. The classical remedy is a universal standard agreed ahead of time: everyone adopts one vocabulary, and the reconciliation is done once, by committee, before the systems ever meet. That remedy is expensive, slow, and permanently behind the systems it tries to govern.

This programme investigates the alternative that becomes available once the systems on each side can **reason**. Two such systems need not wait for a standard; they can reconcile their models **ad hoc** — between themselves, for the task at hand, machine-to-machine. The question is whether that actually works, how far it reaches, and what — if anything — a small published aid adds to it. The answer is built empirically, in a harness where the reconciling parties are language-model agents and every outcome is scored against a validated gold standard, so that the claims rest on measurement across conditions rather than a single worked demonstration.

2. The lift: from a data model to a portable semantic model

The move the whole programme turns on is the **lift**, and it is worth drawing before anything else (Figure A).

A system usually holds its content as a **data model** — a schema and its records as they sit in a database or a controller. That content is meaningful to whoever wrote the schema, but it carries no explicit meaning of its own: a field named `och_grade` with values `1`, `2`, `3` means something to its author and nothing to a reader who was not told. The **lift** is the move from that to a **semantic model**: the same elements given grounded, explicit meaning — a **lexicon** (a preferred label and synonyms), an **ontology** (each term's kind, and its relations to the others), **definitions** and **worked examples**, the concrete **instances** read as evidence of meaning rather than as mere records, and — optionally — a **binding** to a shared reference. Who performs the lift is itself a variable of the study: a live system lifts and explains itself; an inert one is lifted by whatever cognition reads it.

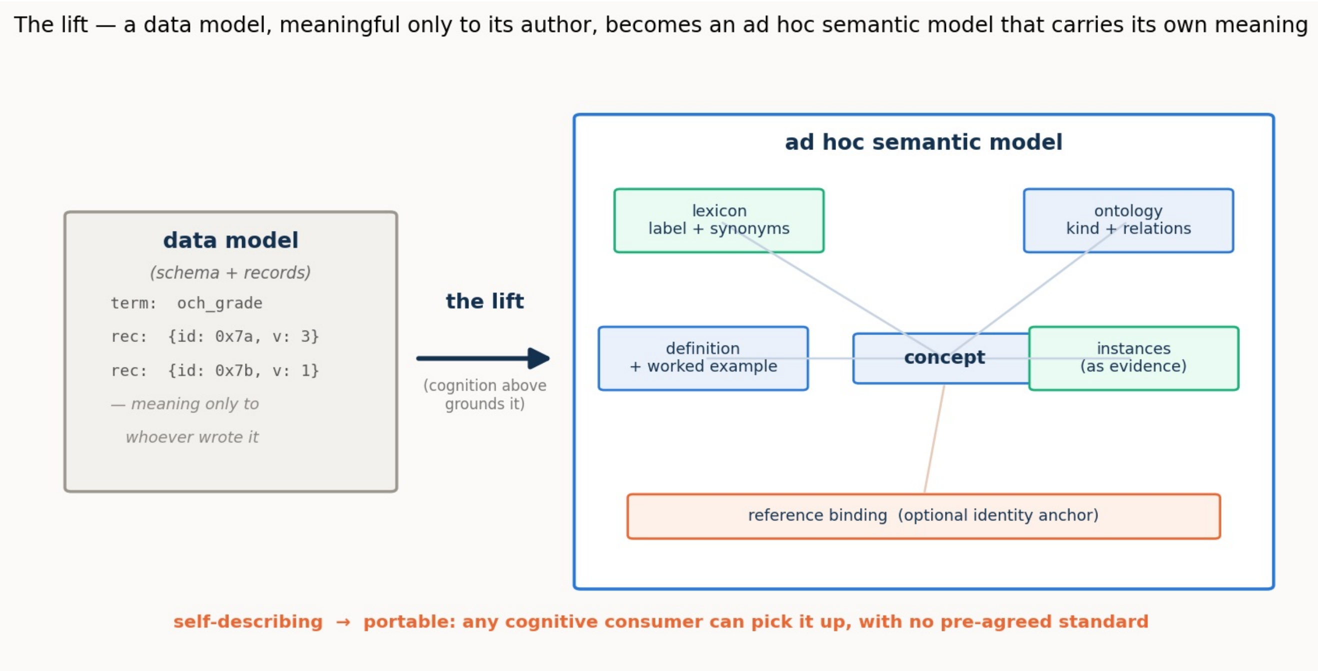

Figure A. The lift. A data model (schema plus records), meaningful only to its author, becomes an ad hoc semantic model whose every element carries explicit meaning — lexicon, ontology, definitions and examples, instances as evidence, an optional identity anchor. Being self-describing, it is portable: any cognitive consumer can pick it up and understand it, with no pre-agreed standard.

The payload of the picture is the word **portable**. A lifted semantic model is **self-describing**, and being self-describing is exactly what lets *any* cognitive consumer — this agent, another agent, a human — pick it up and understand it without a prior agreement. That portability is not a convenience; it is the precondition for the whole approach. Cognitive consumption of a model — using it, reasoning over it, reconciling it with another — requires meaning made explicit, and the lift is how meaning is made explicit on demand rather than by standardisation. Reconciliation is one instance of cognitive consumption; the settings that follow will also refine an intent against a catalogue, and read an anomaly for its significance, over the same lifted substrate. Get the lift, and the rest of the programme is operations over portable semantic models.

3. Reconciliation over lifted models: the operations

Given two lifted models, **reconciliation** aligns them: it works out which concept on one side denotes the same thing as which concept on the other, fixes the shared attributes so they cannot be misread, aligns the individuals, and confirms the result (Figure B). It is not one act but a small family of **operations**, and naming them once is what lets the findings later be mapped to a precise *where*:

- **Lift** — data model → semantic model, per side (§2).
- **Reference construction** — find or build the thin shared ground the two sides bind through.
- **Schema binding** — align the concepts themselves: the **lexical** operation (names and synonyms) and the **ontological/structural** one (kinds, relations, decomposition).

- **Attribute pinning** — fix the meaning of a shared field, so a *committed payload* rate is not read as a *line* rate, a *bound* not as a measured value.
- **Instance co-reference** — decide which individual on one side is which on the other (entity resolution over keys, attributes, and topology, sometimes only settled by probing the live system).
- **Verification** — confirm a proposed correspondence, by round-trip, by invariant, by a virtual provision-and-read-back, or — where the relation is a refinement — by a satisfaction check.
- **Pragmatic resolution** — settle what the reconciled thing is *for*: whether a degraded offer is acceptable, whose realm governs a shared field, whether an anomaly warrants a page.
- **Composition** — group the reconciled parts into wholes (symptoms into one incident).
- **Lifecycle recurrence** — re-reconcile as the live situation changes over time.

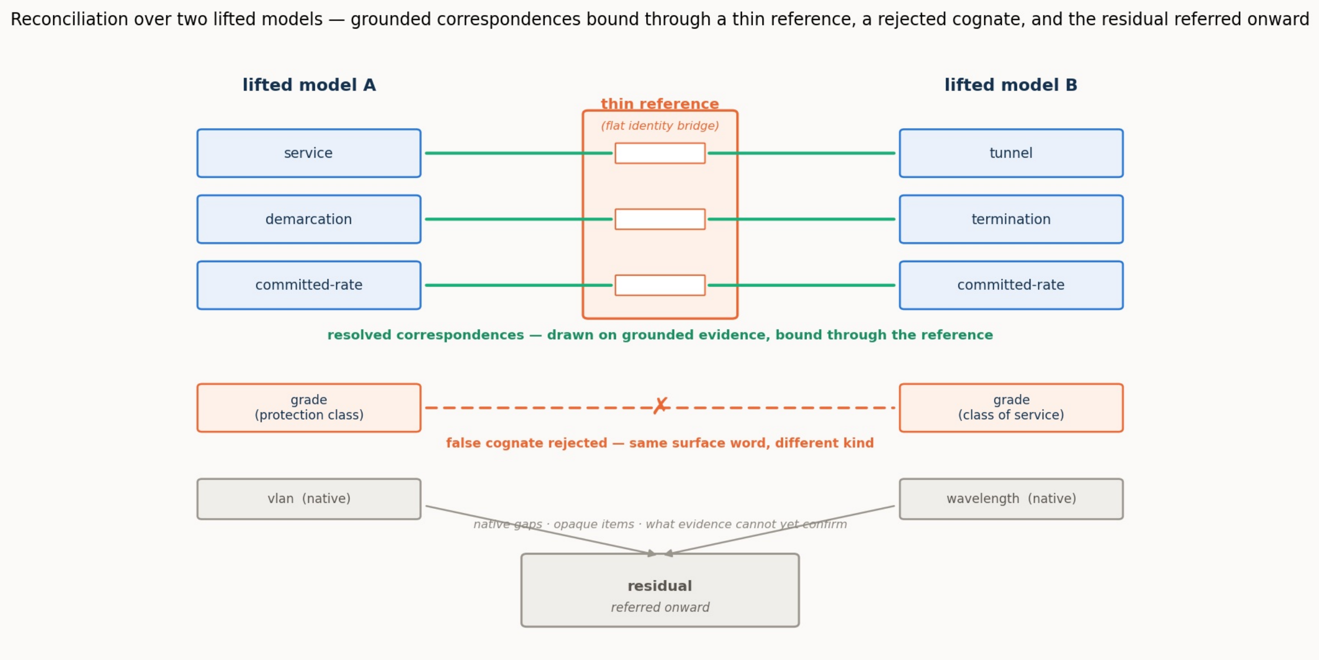

Figure B. Reconciliation over two lifted models. Correspondences are drawn on grounded evidence and bound through a thin reference that acts as a flat identity bridge; a look-alike that shares only a surface word is rejected as a false cognate on its kind, attachment, and instances; and what evidence cannot yet confirm — native gaps, opaque items — is left in the residual, referred onward. The reference carries no structure of its own; it is parasitic on the two grounded models it connects.

Two features of Figure B recur in every setting. The first is the **false cognate**: two concepts that share a surface word but denote different things — an optical *signal-grade* and a commercial *service-grade*, a transport *grade* and an IP *grade*, a legacy *alarm* and an NMOP *anomaly*. Reconciliation must draw the true correspondences *and* refuse the look-alikes, and it refuses them on grounded, non-lexical evidence — kind, attachment, instances — which a purely lexical matcher, keying on the shared label, cannot do. The second is the **residual**: the correspondences a pass does not close. A residual is not an error; it is what a reconciliation honestly *refers onward* rather than guessing — to further machine cognition where the agents are live to continue, or to a person where they are not. How large the residual is, and what drives it, is one of the programme's central measurements.

4. The cognition spectrum, and the residual as a shortfall

The master control across all four settings is **where the cognition sits** — the **cognition spectrum**. It runs from both sides live and interrogable, through one side inert, to both inert:

- **both-cognitive** — each side is a live reasoner, the authority on its own model, able to explain itself and answer questions;
- **one-inert** — one side is a mute snapshot exposing only its structure and instances; the live side

must reconstruct its meaning;
- **both-inert** — neither side can explain itself; a third party reconstructs both from structure and data, and can only *propose* candidates for external adjudication.

A structural fact about this spectrum organises every result that follows. In the fully-cognitive case, resolution of any reconciliation question is total *in principle*, for two reasons that are themselves functions of live cognition on each side. First, the agents can exchange **unbounded further information** — each is the live authority on its own model, so whatever is ambiguous the other can ask about and get an authoritative answer. Second, because reconciliation operates on the models rather than the live network, the agents can **run decisive experiments in virtual space** — provision a candidate through a proposed correspondence, operate it, read it back, and check the invariants. Any question is therefore confirmed, refuted, or authoritatively decided, and nothing need be left unresolved. Both mechanisms are functions of live cognition, so as cognition recedes they fall away: with one side inert the live agent can probe but not interrogate or co-design an experiment; with both inert there is no one to ask and no joint experiment to run. The **residual** a reconciliation must leave and refer onward is, in exactly this sense, the **shortfall from full cognition's reach** — and it grows as that reach recedes. It is not a fixed floor the fully-cognitive case merely reaches; between two fully-cognitive agents there is, in principle, none.

This is the frame inside which every finding sits, and it carries the programme's central claim. Lexical and descriptor methods carry a reconciliation to roughly ninety percent — names, then names plus a gloss — and there such methods have historically stopped, the remainder left to an agreed standard or to human judgement. What the spectrum shows is that the remainder need not wait for either: where both systems can reason, the reconciliation completes **autonomously**, and only as cognition recedes does closing the gap fall back to a reference or a person. **It is cognition that completes a reconciliation**, and the cognition spectrum is the measure of how far the automation reaches before it must hand off.

5. The thin reference: substitute, supply, and the limit at authority

Against that frame, a thin, published **shared reference** is switched on and off at each point on the spectrum — as much a probe of what cognition was doing as an object of study in its own right. The reference is deliberately small: a flat set of entries, each an identity anchor plus a few descriptive fields (a label and synonyms, a shallow class, a one-line definition, a canonical example). It has no ontology of its own — giving it one would turn it back into the universal standard the approach exists to avoid — and its power is coordination, not content: two systems that bind to the same entry thereby denote the same thing. It is **parasitic** on the two grounded models it connects, which is precisely why it can be so thin.

What such a reference does, across the settings, resolves into three distinct roles, and keeping them apart is essential:

- It can **substitute for cognition**. For a capable agent on a lexical/structural task, the reference does the reasoning's work — collapsing the agent's hidden deliberation by one to two orders of magnitude and yielding a perfect, verified reconciliation.
- It can **supply missing information**. Where a side is inert, the reference can hand the reading agent facts it can no longer obtain by interrogation — a committed guarantee, a unit, a categorical definition — restoring the verification the missing probe would otherwise have done.
- It **cannot supply authority**. Where what is missing is not a fact but a *judgement* — whether a degraded offer is acceptable, whose realm governs a contested field — no reference substitutes. A reference can tell you what a service guarantees; it cannot tell you whether this customer will accept it. Where the gap is information, a thin reference closes it; where the gap is authority, only cognition — live, or pre-placed as a policy — closes it.

The reference's value is also **capability-gated** and distributed unevenly along the spectrum. Its effort benefit peaks with strong, live cognition — a weak agent may not be able to exploit it, and an inert side can turn the extra material into a burden. Its error-prevention benefit runs the other way, concentrating where cognition, and therefore verification, is weakest. One anchor, opposite roles, and — a finding the settings sharpen — a definite **anatomy**: not every thin reference is equal, and the cheapest useful one is not the cheapest possible one.

6. The instrument

Every setting is measured in the same harness. A **case** is two lifted semantic models plus a gold-standard reconciliation *derived from the models by a script and validated* before any run, so the gold cannot drift from the models it scores. A **reasoning stack** stands in for the reconciling cognition — in the fully-cognitive case, an exchange between two live agents; with a side inert, the live agent reconstructing the mute one; with both inert, a third party — and the harness scores the stack's output against the gold and records both quality and cognitive effort. Quality is measured by **precision** (of what it proposes, the fraction correct), the **resolved fraction** (of the true correspondences, the fraction it actually commits, the rest referred to the residual), **surviving false cognates** (planted traps taken), and the **residual** itself; effort, for a language-model stack, by **reasoning tokens** (hidden deliberation), total tokens, and latency. Throughout, a resolved fraction below one is not by itself a failure — it is reach deliberately traded for honesty, the unconfirmed correspondences referred onward rather than guessed.

The reconciling agents are run over one **model ladder** for the whole programme — three models spanning a capability range, `gpt-5.6-sol` (**strong**), `gpt-5-mini` (**mid**), and `gpt-5-nano` (**weak**), referred to as **sol**, **mini**, and **nano** — chosen so the ladder isolates model strength rather than provider or architecture. The programme's hypotheses are the ones the first setting states and the others inherit: that cognitive agents can reconcile divergent models ad hoc (the concept holds); that the placement of cognition governs what a reconciliation achieves, costs, and can verify; that a thin reference substitutes for cognition, capability-dependently, and prevents errors where cognition is weakest; and that reconciliation work scales linearly with a shared reference against quadratically without one. What each setting adds is a new operation brought under test — verification and instances, then pragmatics as movable policy, then the standard-free bind, then the pragmatic frontier of significance — against the same frame and the same instrument.

Part II — The four settings

Each setting takes the same frame to a new operation. The sections below are distilled to a common rhythm — what is new against the first setting, the case in one paragraph, the operations put under test, what was proven, and what it means — and each points to its own report for the full evidence.

7. Setting 1 — Configuration: two standard models of one network

Full report: 1/4 · Configuration.

What it establishes. This is the founding setting: it puts the whole idea to its first empirical test and fixes the frame the others inherit. Two independently authored **public standard** models of one optical transport network — ONF **TAPI** on one side, IETF **TEAS/ACTN** on the other — describe the same nodes, links, and services, each in its own vocabulary. A TAPI *connectivity-service* is a TEAS *tunnel*; a TAPI *link-termination-point* looks like a TEAS *tunnel-termination-point* but is a different thing — a link end, not a trail head. Because both models are public standards the agents recognise, cognition can lean on that recognition to bridge them, which makes this the cleanest place to isolate the mechanism.

Operations under test. The reconciliation as an **equivalence** (this term *is* that term): the **lexical** and **ontological/structural** binding, **verification** run as its own step, and **instance co-reference** — with the **pragmatic** operation deliberately held fixed, deferred to the later settings. The thin **reference** is the lexical one, switched on and off across the spectrum.

What was proven. The concept holds, measured across conditions rather than shown once: cognitive agents reconcile the two models correctly and ad hoc. Within that, the reference **substitutes for cognition** — for the strong agent it collapses hidden deliberation by one to two orders of magnitude and yields a perfect, verified reconciliation (precision and resolved fraction of one) across the whole spectrum. The benefit is **capability-dependent** and not monotone: for a weak agent the reference can *add* effort when a side is inert, and the naive "reference helps more as cognition recedes" hypothesis is refuted *for effort* even as it holds *for correctness* — precision separates the models on the hard case (strong 0.94→1.00, mid 0.88→0.96, weak 0.91→0.97 with the reference) and the reference prevents the errors a weak agent otherwise commits. **Verification** catches the false cognates that slip the proposal, driving precision toward one across the spectrum; but without the reference this catching *lowers* the resolved fraction, because correct correspondences a receding cognition can no longer confirm are referred onward rather than asserted — and the reference restores exactly that confirmation. Reconciliation work scales **linearly** with a shared reference against **quadratically** without one (verified by construction to N = 12). And two folded sub-studies carry the point past the schema terms: at the **instance** level the resolvability of same-looking individuals is budget-limited at full cognition (driven to a full resolve with more probing) but becomes **structural** once a side goes inert (no budget helps, because the inert side cannot be interrogated); and a **pre-lift baseline** confirms the lift itself is the lever — matching on the bare lexical surface leaves a quarter to a third of the correspondences unfound (resolved fraction 0.66–0.76), and restoring the lifted content recovers them (0.92–0.97).

What it means. The founding claim is in hand: cognition completes the reconciliation, the reference substitutes for it and prevents its errors, and the spectrum governs both. Everything after this builds on that frame; the one operation held fixed here — the pragmatic — is the thread the remaining three settings pick up.

8. Setting 2 — Intent: refinement, negotiation, and a service that renegotiates itself

Full report: 2/4 · Intent.

What is new. The first setting reconciled by **equivalence**; this one reconciles a declarative **intent** against a concrete **realisation** by **refinement**. "Latency below 5 ms" is not equal to any service on offer — it is a *bound* that a service either clears or does not, and many services may clear it. Once the relation is refinement, everything the first setting held fixed comes into play at once.

The case. A customer's agent **O** holds an intent — a New York–Frankfurt connection, ≥ 8 Gbit/s, ≤ 5 ms, four-nines, protection not required — every clause a bound. An operator's agent **N** holds a catalogue of concrete optical services, each with a real bandwidth, latency, availability, protection scheme, and cost. Neither speaks the other's language; they must work out which realisation *satisfies* the wish.

Operations under test. Verification becomes a **satisfaction** check (you cannot round-trip a lossy refinement, so you test the chosen realisation against every bound); a genuine two-sided **negotiation** appears when no service meets every bound (N computes a best-achievable offer, O must decide accept or reject); the **pragmatic** operation enters for the first time, carried in a small, portable **movable policy** the customer holds — a priority ordering, hard bounds, an affordability floor, a flow-class rule; and the whole exchange **recurs across the service's life**.

What was proven. The central insight, now measured on a real negotiation: with both agents live, the reconciliation — negotiation included — completes autonomously (decision accuracy 1.0 for sol and mini), with no human in the loop. The spectrum gains a new rung that turns out to be the sharpest result:

remove the customer's live judgement, and a **pre-placed movable policy** still closes the decisions it was authorised for (1.0), while a **mute** customer with no policy cannot — accuracy falls to zero and the agents correctly **refer all three** decisions onward rather than guess. The pre-placed policy is a concrete mechanism for **pushing the hand-off boundary**: the customer's cognition, placed in a portable artefact ahead of time, closes autonomously what a mute description must refer. The second finding draws the reference's limit: where a side is inert, a published reference *supplies information* (the strong agent, blind at both-inert, refuses to affirm satisfaction and scores 0.29; publish the invariant guarantee floor and it has an anchor to check against, rising to 0.71) — but it *cannot supply authority*: at both-inert no reference moves the negotiation, because what is missing is the customer's judgement itself. The capability gradient is steep (to reach its decisions the strong agent spent ~150 reasoning tokens, the mid ~860, the weak ~5,400 — thirty-five times the strong agent's effort, for lower accuracy), and a four-hop **lifecycle** — bought, self-healed, referred, restored — is walked correctly end to end by the strong model (hop accuracy sol 1.0, mini 0.88, nano 0.62).

What it means. Cognition completes even a negotiation, and the pragmatic operation is now a measured axis rather than a fixed backdrop. Information has a published stand-in; authority does not — and the pre-placed policy shows how to carry a party's authority to where a person would otherwise have to stand.

9. Setting 3 — Cross-domain: reconciling with no public standard

Full report: 3/4 · Cross-domain.

What is new. The deliberate complement to the first setting. There, two *different* models faced each other — but both were **public standards** the agents already knew, so cognition could lean on recognition. Here that crutch is gone: both models are **home-grown and private**, recognisable to no one in advance. The operations are the first setting's — lift, bind, a thin reference — but with the standard removed, the *execution* diverges, and that divergence is the subject.

The case (Figure C). Meridian, a home-grown transport OSS, thinks in circuits, bearers, wavelengths, protection grades. **Cascade**, a home-grown IP/VPN controller, thinks in services, attachments, VLANs, classes of service. Their worlds touch at exactly one seam: a Cascade service must ride a Meridian circuit as its underlay. One order across that seam needs five bindings — circuit↔underlay and hand-off↔attachment (the same object under two names), and three requirements pinned (a *committed* rate not a line rate, a latency *bound*, protection against a *path* failure) — and one look-alike refused: both models carry a **grade**, but Meridian's is a transport protection class and Cascade's an IP class of service. Same word, unrelated meanings, and no standard to consult.

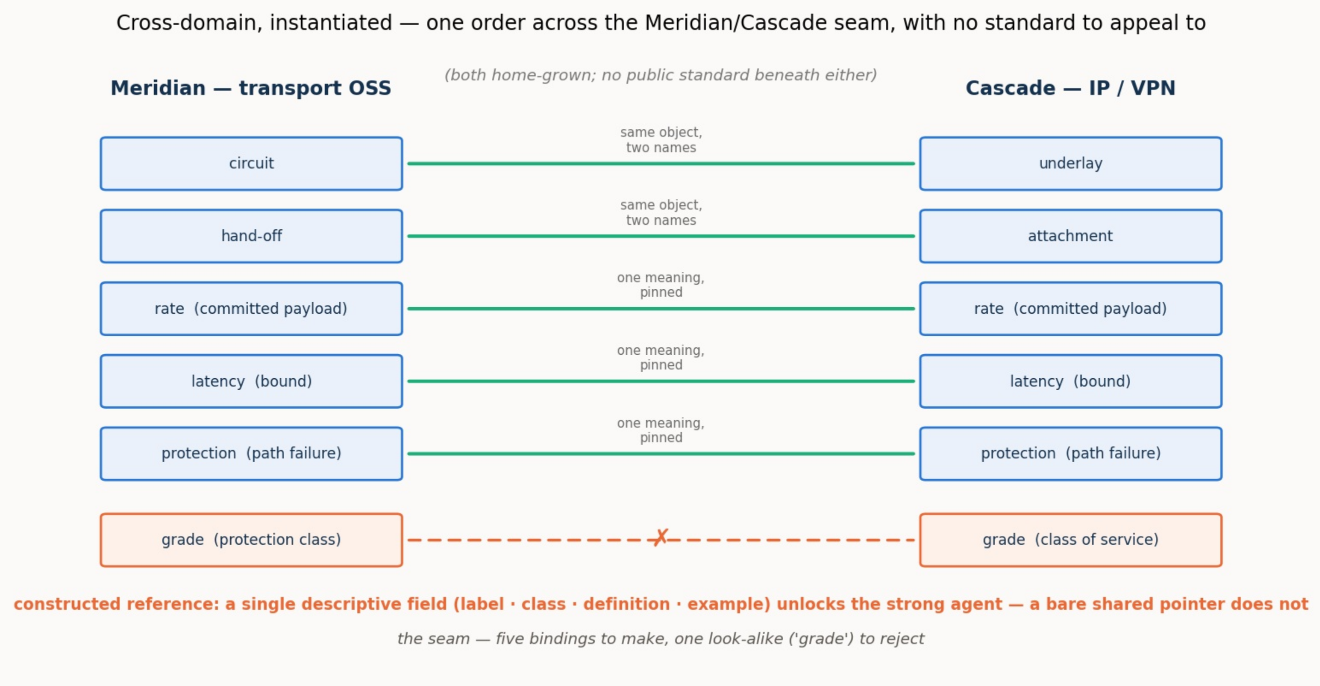

Figure C. One order across the Meridian/Cascade seam. Five bindings to make — two renamings and three pins — and one false cognate, "grade", to reject, across two private vocabularies with nothing public beneath them. A constructed reference supplies the shared ground; a single descriptive field in it is enough to unlock a capable agent.

Operations under test. A single, early **schema-binding** pass — deliberately isolated from the rest of the process — measured across the spectrum with and without the reference the two agents would themselves **construct**; plus a **reference-field ablation** and the **pragmatic** operation of authority attribution (whose realm owns each shared field). The instance operation is bracketed, on the argued grounds that it reproduces the first setting's result.

What was proven. The central result is a **mirror**: with the constructed reference absent, the **strong agent under-commits and the weak agent mis-commits**. The strong agent will not guess across two foreign vocabularies — it binds only the names that already coincide (resolved fraction ~0.5 at both-cognitive, ~0.4 once a side is inert) but at perfect precision and with no false cognate taken: this is **omission**, deferral, not error. The weak agent does the opposite — it binds freely (resolved fraction 0.8–1.0) at a precision that never clears 0.83 and falls to 0.57 inert, taking the cross-domain "grade" trap: **commission**, confident error. A single thin thing repays both: constructed, the reference lifts the strong agent to a full close and lifts the weak agent's precision toward one. The **ablation** sharpens what the reference must carry: any *single* descriptive field — a shared label, a class, a definition, or an example — is enough to unlock the strong agent's commitment (each drives it to a near-perfect close), while a **bare shared identifier with no description is worse than nothing** (precision 0.50, below the no-reference floor), because the agent binds by an opaque token and binds wrongly. The reference does not work through the shared *pointer*; it works through the shared *description*, and even the thinnest description suffices. On the **pragmatic** axis, the same boundary the intent setting found appears again: the reference reaches *meaning* but not *authority* — it can pin that a rate is a committed payload, but not whose realm governs it — and a characteristic "transport owns everything it carries" bias marks the weaker models.

What it means. With no public standard beneath two models, cognition still closes — but only once it has built the shared ground, and **building that ground is the work**. The strong agent's low reference-absent numbers are not a limit of cognition; they measure the worth of the one step the isolated pass was not allowed to take. Even a very thin ground suffices for a capable agent, provided it carries meaning and not merely a pointer.

10. Setting 4 — Observability: an alarm is not an anomaly

Full report: 4/4 · Observability.

What is new. The setting where the pragmatic operation moves from the wings to the centre. The descriptor level — what a thing *is* — is settled by a standard, and the entire operational question is one of **significance**: whether an observed deviation matters, and how observations compose. It also carries the programme's deepest false cognate, an **ontological** one.

The case (Figure D). Agent F, a legacy fault manager, emits an **alarm**: one object bundling an event, an undesirable state, a fixed severity, and a static probable-cause. **Agent G**, an IETF **NMOP** agent, speaks the RFC 9940 ladder, which separates what legacy conflates — event, anomaly, symptom, fault, alarm, problem, cause, incident — and annotates each anomaly with anomaly-semantics metadata (a concern score, a confidence score, a plane, a pattern, a lifecycle stage, a season). The worked example is one live anomaly: a pre-FEC bit-error-rate reading on a wavelength begins to rise. In the legacy world it fires an alarm and a page; in the NMOP world nothing is decided yet — whether it warrants a **page** (an alert to an on-call human) depends on its concern and confidence and its context (a maintenance window makes the same deviation expected), and whether it is one incident or many depends on correlating it, across layers, with the symptoms it causes.

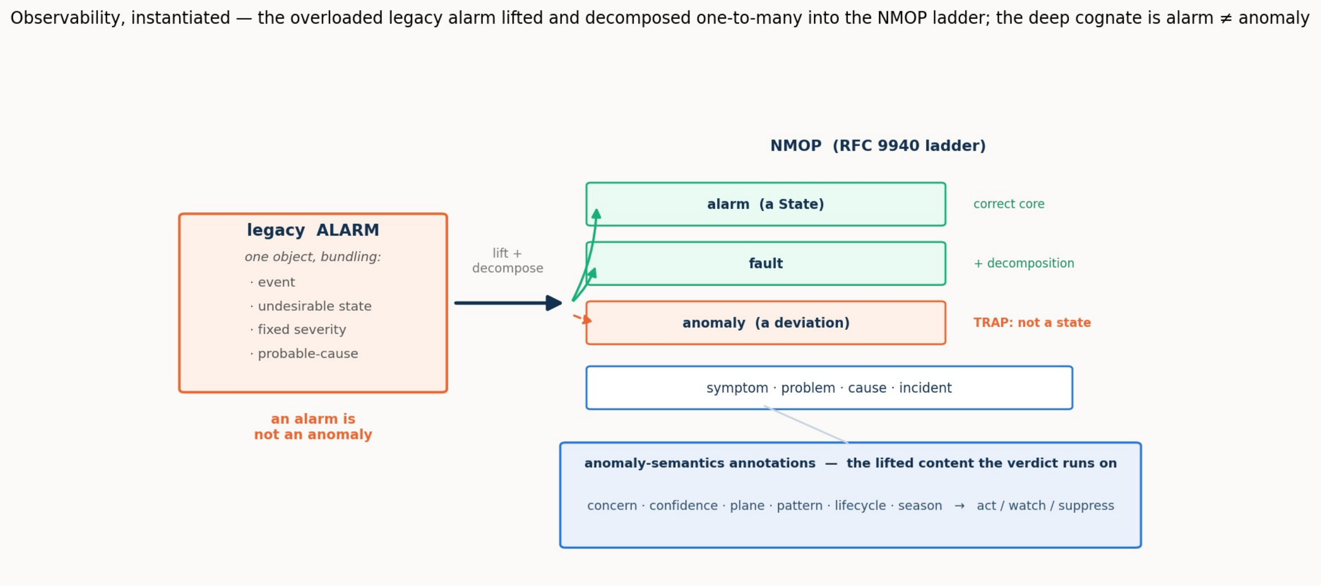

Figure D. The overloaded legacy alarm is lifted and decomposed one-to-many into the NMOP ladder: its correct core is the NMOP alarm-State (and the fault it implies), while the trap is the anomaly — an alarm is a State, an anomaly is a deviation, and conflating them is a category error, not a mislabel. The anomaly-semantics annotations are the lifted content the significance verdict runs on.

Operations under test. Two acts. **Act 1** is a **schema binding** carrying the ontological false cognate (alarm↔anomaly, which no structural cue separates) and a **one-to-many decomposition** (the legacy alarm maps to both the NMOP alarm-State and the fault). **Act 2** is pure **pragmatic resolution and composition**, each run with the anomaly-semantics **ON** and **OFF**: a **verdict** task (act / watch / suppress for each anomaly under a context) and a **correlation** task (group cross-layer symptoms into incidents and name each cause).

What was proven. Three results complete the arc. First, the **ontological cognate is a clean three-rung capability gradient**: the strong agent never conflates alarm with anomaly (with or without the reference); the mid agent conflates them once a side is inert, and the RFC 9940-anchored reference **rescues it completely** (the cognate vanishes, precision returns to 1.0); the weak agent conflates them with or without the reference — beyond rescue. The lexicon pins the ontology for the middle of the ladder, not the bottom. (The honest hard edge of Act 1 is the decomposition: even the strong agent tends to map the alarm to the alarm-State but miss the fault constituent, so the resolved fraction sits near 0.75.) Second, in the verdict task the **pragmatics carry the operative meaning** — with the semantics ON the strong and mid agents suppress correctly during maintenance and the false-page storm disappears (verdict accuracy 1.0 and 0.83, essentially no false pages); with them OFF the legacy pipeline pages nearly everything (accuracy 0.17 and 0.08, roughly four and three-and-a-half false pages) — **but the payoff is capability-gated**: handed the identical annotations, the weak agent barely moves (ON 0.53 vs OFF 0.50). Third, **correlation behaves differently**: given the resource-dependency map, *every* model — the weak one included — folds an optical degradation and the IP loss it causes into one correctly rooted incident (perfect partition ON), and without that map every model fails. The difference is the lesson: correlation's pragmatic is a **structural input** and even a weak agent applies it; the verdict's pragmatic demands **judgement**, and there capability decides.

What it means. Meaning (what an anomaly *is*, pinned by the reference) and significance (whether it warrants a page and how it composes, carried by the pragmatics) are **separable, both necessary, and each gated in its own way**. The pragmatic frontier the descriptor methods never reach is real and decisive — and itself bounded by the agent's capability. The through-line reaches its end: cognition completes the reconciliation; a thin reference supplies the information it needs and stops at what it does not; and the pragmatic layer is the frontier, decisive for meaning and gated by capability.

Part III — Synthesis

11. What works, and how far

Read as one result, the four settings say first the plain thing: **it works, largely as laid out, barring a few surprises.** Two cognitive agents reconcile independently authored, divergent models correctly and ad hoc — completing an equivalence between two standard models (setting 1), refining and negotiating an intent against a catalogue (setting 2), binding across a private domain boundary once they have built the shared ground (setting 3), and reading an observability world for what its signals mean (setting 4) — measured against a validated gold across the cognition spectrum, not shown once by hand.

And it says, more sharply, **where** it works without help. At the fully-cognitive end of the spectrum, the reconciliation completes **autonomously in every setting** — no standard agreed in advance, no human in the loop — including the two operations that look least automatable: a two-sided negotiation (setting 2, decision accuracy 1.0 with both sides live) and a significance verdict (setting 4, accuracy 1.0 with the pragmatics on). The fully-cognitive end is, across all four settings, the automatable end. That is the headline, and everything else in this synthesis is a qualification of it: how the automation degrades as cognition recedes, what a thin reference buys back, and where a capability floor cuts the whole thing off.

12. The findings, as six theses — and mapped to where they act

The results gather into six theses. Each is stated once here and then located, in the two tables that follow, against the **operation** it concerns and the **settings** that evidence it — so that when a finding says cognition (or a reference, or a pragmatic) matters "here," the *here* is a named stage of the process, not a vague gesture.

Thesis 1 — Cognition completes a reconciliation; descriptor methods reach a ceiling and stop. Lexical and descriptor matching carry a reconciliation to roughly ninety percent (setting 1's baseline: lexical surface 0.66–0.76, the lift recovering it to the mid-nineties); the remainder, historically left to a standard or a person, is closed by live cognition instead. This is the programme's spine, and it holds at every operation the settings put under test.

Thesis 2 — The placement of cognition is the master variable, and the residual is its shortfall. What a reconciliation can achieve, cost, and verify is governed by where the cognition sits. Between two live agents, resolution is complete in principle — unbounded interrogation and decisive virtual experiment — so the residual is, in principle, none; as a side goes inert those mechanisms fall away and a residual appears, precisely the shortfall from full cognition's reach. The same story holds at the schema level (setting 1), the instance level (setting 1's budget-limited-becomes-structural curve), the negotiation (setting 2), and the standard-free bind (setting 3).

Thesis 3 — A thin reference substitutes for cognition and supplies information, but stops at authority. For a capable agent on a lexical/structural task it substitutes for the reasoning (setting 1, effort down one-to-two orders, a perfect verified close); where a side is inert it supplies the facts interrogation no longer can (setting 2, sol 0.29→0.71 with the invariant floor; setting 4, the RFC 9940 reference rescuing the mid agent's ontology); but where the missing ingredient is judgement rather than fact, no reference moves it (setting 2's authority gap, setting 3's authority attribution). Information has a published stand-in; authority does not.

Thesis 4 — Failure modes are capability-signed: strong agents omit, weak agents commit. Denied the ground it needs, a strong agent **defers** — it leaves the unresolved in the residual at perfect precision (setting 3's under-commitment; setting 2's strong agent refusing to affirm what it cannot verify). A weak agent **commits** — it binds freely and wrongly, takes the false cognate, and is no less confident on a wrong answer than a right one (setting 3's mis-commitment; setting 1's traps surviving for the weak agent). The strong agent's error is a larger residual; the weak agent's is lower precision. This is why a reference's *correctness* value concentrates where cognition is weakest — it disciplines the committing agent — even as its *effort* value peaks where cognition is strongest.

Thesis 5 — The pragmatic layer is the frontier the descriptor methods never reach — decisive for meaning, and itself gated by capability. What a reconciled thing is *for* — whether a degraded offer is acceptable (setting 2), whose realm owns a field (setting 3), whether an anomaly warrants a page (setting 4) — is where the operative meaning lives, and it is beyond the reach of names, glosses, and references. It is carried by cognition (or by cognition pre-placed as a policy), and its payoff is realised only by an agent strong enough to carry it: handed identical annotations, the weak agent still cannot produce the verdict (setting 4). The one exception proves the rule — where a pragmatic is delivered as a **structural input** rather than a judgement (setting 4's correlation dependency map), even a weak agent applies it.

Thesis 6 — With a shared reference, reconciliation scales; and the reference has an anatomy. Work grows linearly in the number of systems with a shared reference against quadratically without one (setting 1, to N = 12), so the reference's advantage compounds with scale independent of any per-reconciliation effect. And not every thin reference is equal: a single descriptive field unlocks a capable agent, but a bare identifier with no description is worse than nothing (setting 3), and a shallow class tag can actively mislead the weak agent it was meant to help (setting 1's anatomy). A reference is a safety rail carrying *meaning*, not a pointer and not a payload.

Table 1 — the operations of a reconciliation, and where each was put under test. The rows are the process stages of §3; a cell says how the setting exercised that operation, or "—" where it did not. This is the map for locating any finding: the *here* of "cognition matters here" is a row.

operation (the *where*)	1 · Configuration	2 · Intent	3 · Cross-domain	4 · Observability
Lift	founds it; the lift is the lever (0.66–0.76 → 0.92–0.97)	reused	reused; both sides private	alarm lifted and **decomposed**
Reference construction	given (lexical)	given (unit / invariant)	**constructed** — the pivotal act	given (RFC 9940-anchored)
Schema binding (lexical, ontological)	equivalence; reference **substitutes**	shared ground for satisfaction	**the measured stage**: omission vs commission	**ontological** cognate; 3-rung gradient
Attribute pinning	—	bound vs measured metric	committed vs line rate	severity / scores
Instance co-reference	budget-limited → **structural**	same gradient (endpoints)	bracketed (reproduces s.1)	bracketed (reproduces s.1)
Verification	own step; catches cognates, holds resolved fraction	by **satisfaction**	downstream of the isolated pass	verdict stands in
Pragmatic resolution	held fixed (deferred)	movable **policy**; authority ≠ information	**authority** attribution; reference reaches meaning only	**verdict** carries operative meaning; capability-gated
Composition / correlation	—	—	—	robust with the dependency map (all models)
Lifecycle recurrence	—	four-hop loop (self-heal, refer, restore)	—	—
Scaling	linear vs quadratic (N = 12)	—	—	—

Table 2 — the cognition spectrum, and what happens at each placement. The rows are the master variable; the columns say what the reconciliation can do, what it leaves in the residual, and what a reference can buy — consolidated across all four settings.

placement	what the reconciliation can do	the residual	what a reference buys
both-cognitive	completes **autonomously** in every setting — bind, negotiate, verdict	none, in principle	substitutes for cognition (effort ↓ 1–2 orders); little correctness room left
one-inert	live side reconstructs the mute side; probes gone, so some resolutions turn **structural**	grows — the shortfall appears	supplies missing **information**; rescues the mid agent (ontology, satisfaction)
both-inert	third party can only **propose** for adjudication; no interrogation, no experiment	largest; every judgement referred onward	supplies information, **not authority** — cannot move a judgement
pre-placed policy (intent)	closes the decisions the policy was **authorised** for — pushes the hand-off boundary outward	only the un-authorised judgements	policy carries the **authority**; a reference still cannot

13. The surprises

Four results ran against the naive expectation, and they are worth stating as findings rather than smoothing away.

A strong agent scoring *lower* is often a strong agent behaving *better*. Blind at both-inert, the strong agent refuses to affirm what it cannot verify and scores below the mid agent (setting 2's satisfaction 0.29 vs 0.71; setting 3's under-commitment). This is not weakness; it is the honest deferral of Thesis 4, and it is exactly the behaviour a published reference or a live probe converts into a confident, correct close. A raw accuracy number, read without the precision beside it, misjudges it.

A bare shared pointer is worse than no reference at all. One might expect any shared anchor to help; setting 3's ablation shows an opaque identifier with no description dropping precision below the no-reference floor (0.50), because the agent binds by the token and binds wrongly. The reference works through shared *description*, never through the pointer — and the corollary (setting 1) is that a shallow *class* tag, the thinnest description, can mislead the weak agent it was meant to help.

The richer content that raises a weak agent's reach also raises its exposure. The lifted content and instances that let an agent find more correspondences also hand a weak agent more surface to misfire on — its instances turn toxic, over-read as evidence of identity (setting 1's baseline). The lift is unambiguously the lever and unambiguously safe for the strong agent; for the weak agent it is double-edged.

Two pragmatics of the same setting place opposite demands on the agent. In observability, the verdict (a judgement over concern, confidence, and context) is sharply capability-gated, while the correlation (an application of a dependency graph) is robust across the whole ladder. Both are pragmatics and both are decisive, but one demands judgement and one is a structural input — and that distinction, not the label "pragmatic," predicts whether a weak agent can do it.

14. Scope, threats, and what remains

The claims here are **existential and mechanistic** — *this is how ad hoc reconciliation works, and here it is working* — not population estimates. Each setting rests on a single seeded case, built to exercise each mechanism and prove each trap rather than sampled from a distribution, with a small number of trials; the reported patterns are the ones stable across the model ladder and the treatment toggles, and the numbers are indicative rather than tight. The model ladder is three points spanning a capability range, so the *shape* of the gradient between the points is not resolved. Golds are derived from the models and validated, which removes drift but leaves the modelling choices — including setting 4's verdict thresholds, stated openly — as authored rather than found. Public standards may have been seen in training, which could flatter the no-reference conditions; the effect relied on throughout is the *difference* a treatment makes under identical inputs.

Two scoping decisions bound the reading of specific settings. Setting 3 measures a **single-pass schema binding** with the reference-construction step bracketed, so its reference-absent numbers measure the worth of that one step, not a limit of cognition; a direct test of the full **construct-then-bind** protocol — the agents building the reference themselves and then closing, with none pre-given — is the clean way to confirm the thesis in the hardest setting, and is left to further work. And **instance-level co-reference** is measured in full only in setting 1; the later settings bracket it on the argued grounds that it reproduces that result. Larger and more varied cases, more rungs on the ladder, and the end-to-end construct-then-bind run are the natural next steps.

15. In one paragraph

Two systems that can reason do not need a standard agreed in advance to work together; they can lift their data into portable, self-describing semantic models and reconcile those models ad hoc, for the occasion. Across four settings — configuration, intent, cross-domain, and observability — that is what happens: at the fully-cognitive end the reconciliation completes autonomously in every case, negotiations and significance verdicts included, with no standard and no human. Cognition is what completes it; descriptor methods reach a ceiling and stop, and the placement of cognition governs how much of the remainder is closed by machine and how much is honestly referred onward. A thin published reference earns its place inside this frame — substituting for a capable agent's reasoning, supplying an inert side the information it lacks, scaling the work from quadratic to linear — but it reaches information and stops at authority, and it must carry meaning rather than a bare pointer to help at all. And the pragmatic layer — what a reconciled thing is for, whether it matters, who decides — is the frontier the descriptor methods never reach: decisive for meaning, carried by cognition or by cognition pre-placed as a policy, and realised only by an agent capable enough to carry it. That frontier, and the capability floor beneath it, are where the next work lies.